

Supporting Information for *Evaluating
multi-season occupancy models with
autocorrelation fitted to heterogeneous datasets*

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F Evaluating occupancy models with spatially uncorrelated (unstructured) random effects.

Below we describe the multi-season site occupancy model of [Outhwaite *et al.* \(2018\)](#), which we used in our sensitivity analyses. This model represents an improvement from the occupancy model used in [Isaac *et al.* \(2014\)](#) in the sense that it adopts a random walk (normally distributed) prior for the temporal random-effect parameter b_t . This improvement allows the sharing of occupancy information between consecutive years. The model can be described in the following manner respecting the hierarchical structure we used in our main text.

The first part of the model is an occupancy state process, where the latent occupancy state z_{it} of a single species at $i = 1, \dots, I$ sites and $t = 1, \dots, T$ primary occasions is modeled. The state z_{it} is drawn from a Bernoulli distribution depending on the probability of occupancy of the site ψ_{it} . The ψ_{it} is modeled in function of the covariate(s) varying over sites and primary occasions X_{it} , and β is a vector of coefficients including the intercept and the slope(s) (eq. 1). The model writes

$$\begin{aligned} z_{it} | \psi_{it} &\sim \mathcal{B}(\psi_{it}), \\ \text{logit}(\psi_{it}) &= \mathbf{X}_{it}^T \beta + u_i + b_t \end{aligned} \tag{1}$$

with $\mathbf{X}_{it}^T = [1 \ X_{it}]$ to keep with notations in [Doser & Stoudt \(2024\)](#). As the original model formulation has no covariates, this model design is an adaptation with intercept and covariate effects. We also adapted the formulation of the site-level spatially uncorrelated random effects u_i , which was:

$$u_i \sim \begin{cases} \mathcal{N}(0, 10^4) & \text{for } i = 1 \\ \mathcal{N}(u_i, \tau_u) & \text{for } i > 1 \end{cases} \tag{2}$$

where τ_u is the precision parameter of the Normal distribution, calculated as $1/\sigma_u^2$. This parameter controls the variance of occupancy probability across sites. The site level random effects were defined through two Normal distributions, with the first one having an average fixed at zero $\mu = 0$ and variance to $\sigma^2 = 10^4$ in order to improve the identifiability of the intercept. Priors for subsequent sites were determined via the second term in eq. 2. The prior of the standard deviation (used to calculate τ_u) was $\sigma_u \sim U(0, 5)$.

The temporal random effect b_t follows a random walk model defined in eq. 3:

$$b_t \sim \begin{cases} \mathcal{N}(\mu_b, 10^3) & \text{for } t = 1 \\ \mathcal{N}(b_{t-1}, \tau_b) & \text{for } t > 1 \end{cases} \quad (3)$$

where τ_b is the precision parameter of the Normal distribution, calculated as $1/\sigma_b^2$. τ_b controls the variance of occupancy probability in primary occasion t relative to the previous one $t - 1$. The prior for b_1 came from a Normal distribution with $\mu_b = 0$ and variance $\sigma^2 = 10^3$, and the prior average for subsequent primary occasions were determined via the second term in eq. 3. The prior for the standard deviation (used to calculate τ_b) was $\sigma_b \sim U(0, 5)$.

The detection model of Outhwaite *et al.* (2018) accounts for effects of primary occasions – where detection probability is estimated for each primary occasion (year in practical terms) – and species list length per site, year, and secondary occasion (survey effort) on detection probability. Since we want the detection model to be similar to Doser & Stoudt (2024) model, our model writes (eq. 4):

$$y_{itj} | z_{it} \sim \mathcal{B}(z_{it} \times p_{itj}), \quad (4)$$

$$\text{logit}(p_{itj}) = \mathbf{v}_{itj}^T \boldsymbol{\alpha}$$

where α is a vector of coefficients, including the intercept and the effect of the covariate v_{itj} on p_{itj} (eq. 4). The species encounter history y_{itj} is then conditionally related to the latent occupancy z_{it} , meaning that for a truly occupied site and primary occasion $z_{it} = 1$, the species will be detected in one individual secondary occasion with probability p_j (eq. 4). If unoccupied $z_{it} = 0$, then the species cannot be detected.

Prior choice has a strong influence on the performance of the random walk occupancy model (see supplementary material in Outhwaite *et al.* (2018)). The occupancy intercept in the original model formulation had $\beta_0 \sim \mathcal{Be}(\alpha, \beta)$, with α and β controlling the shape of the Beta distribution (not to be confused with model coefficients presented above). In our analyzes, we followed the authors and set both parameters to $\alpha, \beta = 2$. The same prior was used for the detection intercept α_0 . The logit function was then applied to transform a probability into a logit-scale intercept. Priors for regression slopes (from both occupancy and detection models) were taken from a Normal distribution with average and precision as $\mathcal{N}(0, 0.001)$.

This model was fitted to 42 simulated datasets under the design of study 2 - scenario 1: Poisson sampling design, effect of latitude L_i on occupancy, effect of observation-level covariate v_{itj} on detection. The model was written in BUGS language and analyzed in JAGS through the R package jagsUI (Kellner, 2024). Parameter estimates using the simulated datasets were obtained with three parallel MCMC chains of 10,000 iterations each, burn-in of 5,000 iterations, and thinning each 10 iterations (yielding 1,500 posterior distribution draws in total). It took 18 days to run the model for 42 datasets x 16 autocorrelation scenarios per dataset using a Dell Inc.® Precision Tower 3620 with processor Intel(R)Xeon(R)E3-1240 v5x8 with 32GB RAM and 8 cores.

The results for this sensitivity analysis are shown below (Figs. F.1 to F.3) and discussed in the main text.

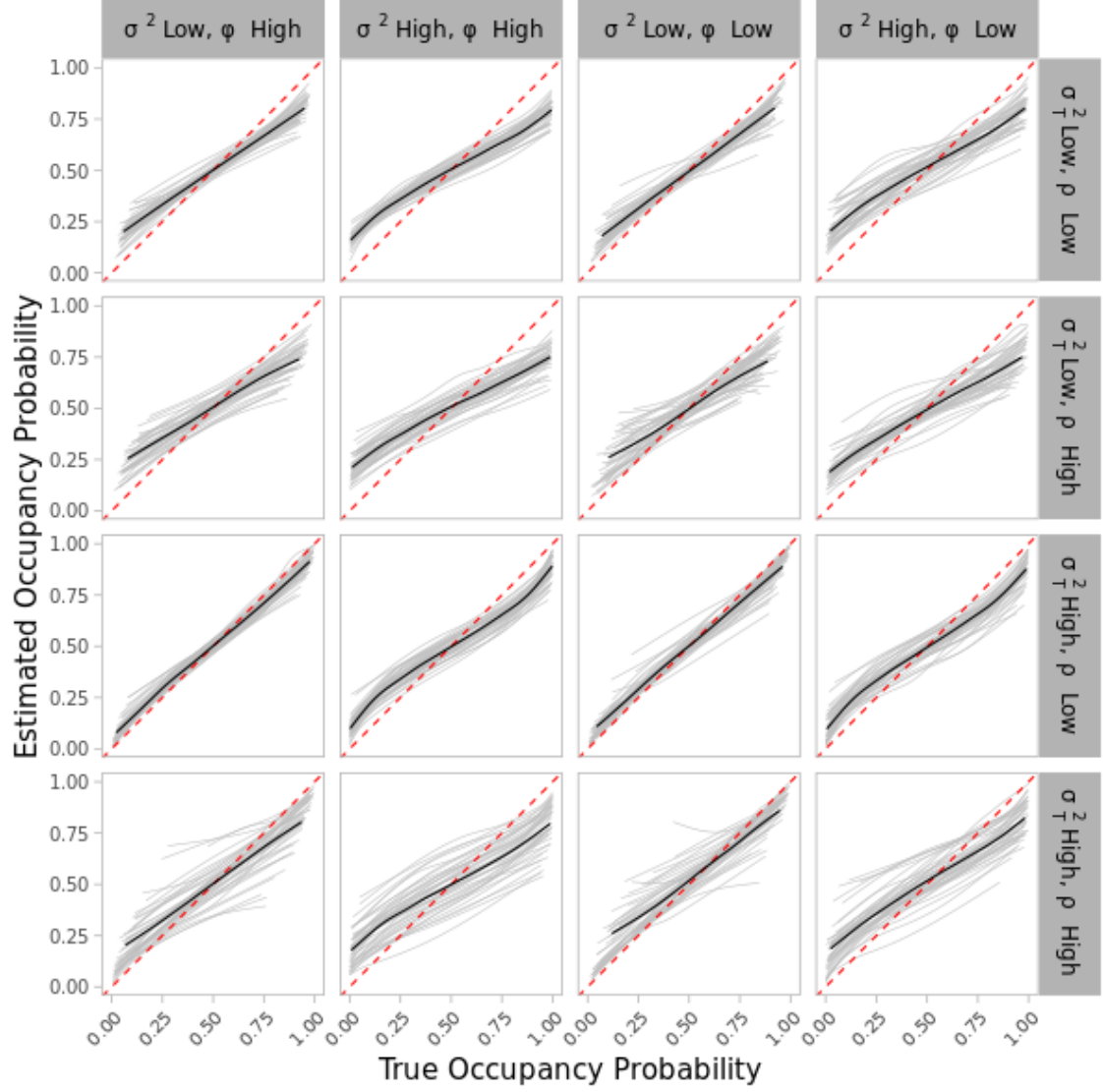


Figure F.1: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis). Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively, with the following levels: high $\phi = 15$, low $\phi = 3.75$, high $\rho = 0.9$, low $\rho = 0.5$, high $\sigma_T^2 = 1.5$, low $\sigma_T^2 = 0.3$, high $\sigma^2 = 1.5$, low $\sigma^2 = 0.3$. These simulations used data from study 1-scenario 1. Each gray line represents the locally estimated scatter plot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 42 simulated datasets, and the red dashed line depicts a 1:1 relationship.

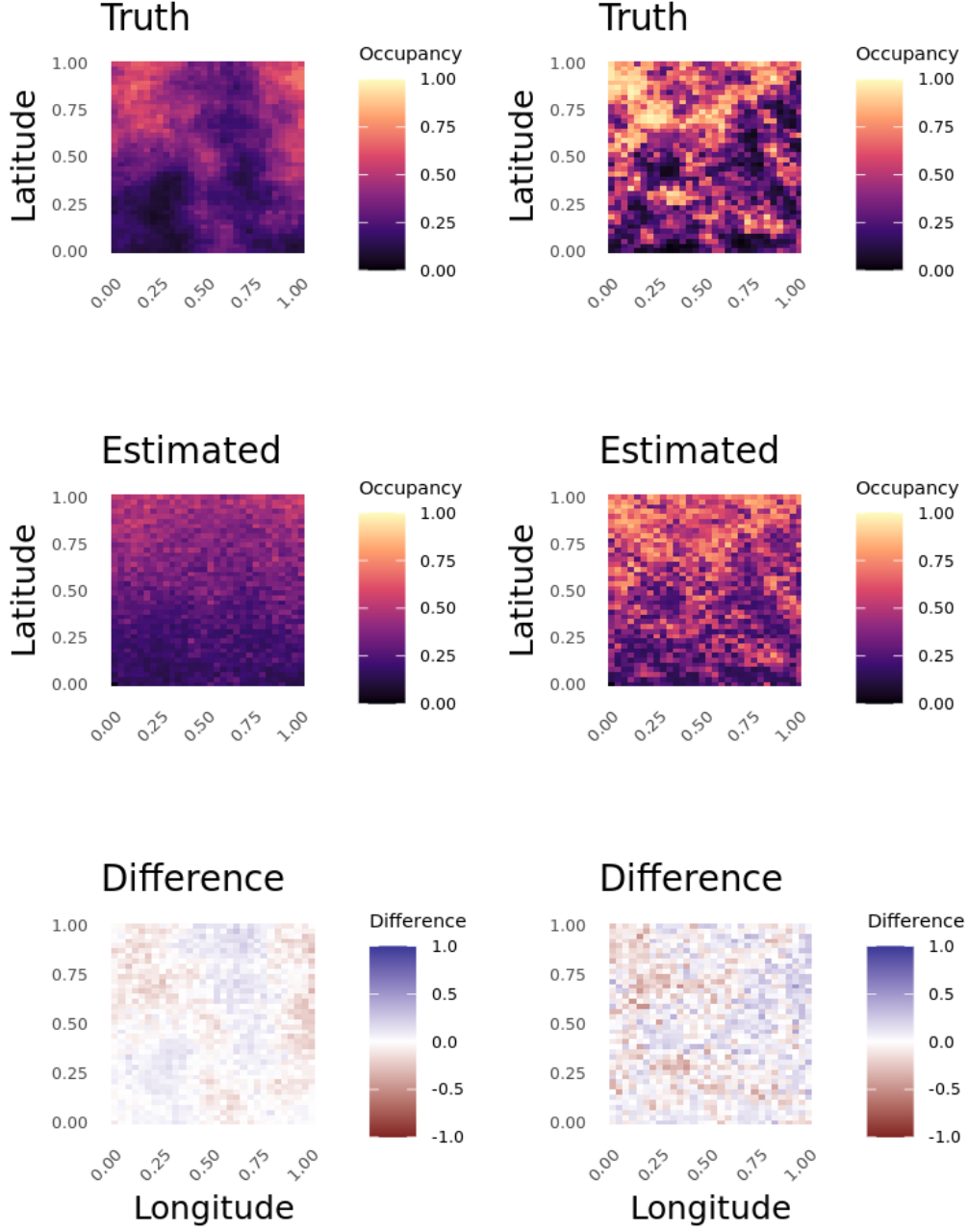


Figure F.2: Spatial map of true site occupancy probability ψ_{it} (top), estimated site occupancy probability $\hat{\psi}_{it}$ (middle), and the difference between them (bottom) across scenarios of high (left) and low spatial and temporal autocorrelation (right). Estimates of $\hat{\psi}_{it}$ were produced by the model with spatially uncorrelated site random effects. Positive differences (in blue) indicate overestimation relative to the true occupancy, whereas negative differences (in red) indicate underestimate relative to the true occupancy. Results for the first simulated dataset and the first primary period of estimation ($t = 1$). Data shown in the left plot were simulated with $\phi = 3.75$, $\sigma^2 = 0.3$, $\rho = 0.5$, $\sigma_T^2 = 0.3$ (strong spatial autocorrelation and weak temporal autocorrelation), whereas in the right the values of these parameters were $\phi = 15$, $\sigma^2 = 1.5$, $\rho = 0.9$, $\sigma_T^2 = 1.5$ (weak spatial autocorrelation and strong temporal autocorrelation).

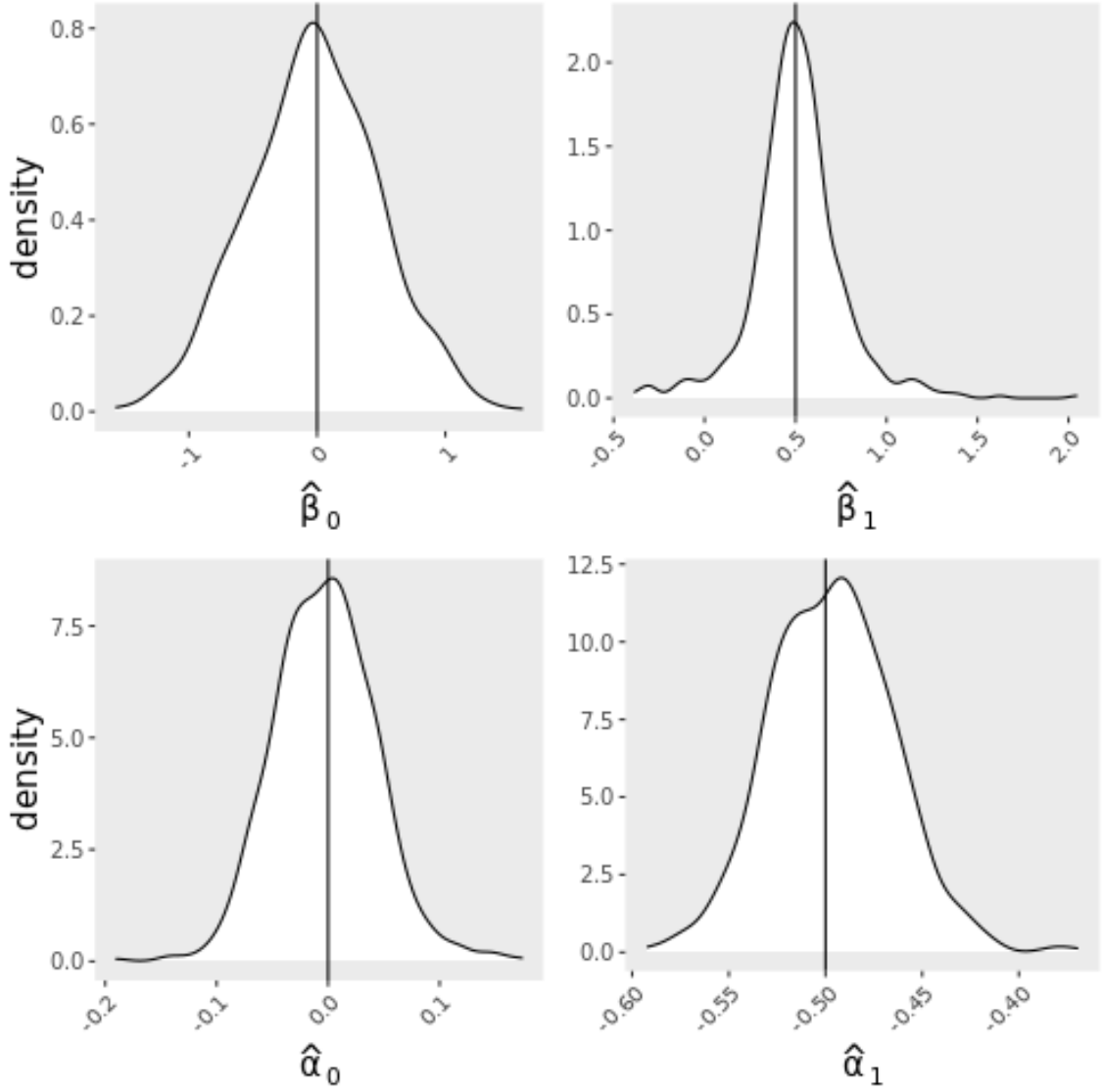


Figure F.3: Density plots showing the model coefficients estimated by the model with spatially uncorrelated random effects. The black vertical line depicts the truth of each parameter. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively. The density was created using the output of the analysis of 42 datasets simulated with $\phi = 15$, $\sigma^2 = 1.5$, $\rho = 0.9$, $\sigma_T^2 = 1.5$.

References

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